

EXP6: Vacuum Cleaner Problem

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Aim:

To develop a Python program that implements a Simple Reflex Agent for the Vacuum Cleaner problem, so that the agent cleans a two-room environment (Room A and Room B) based on its current percept.

Algorithm:

Step 1: Represent the environment as a dictionary with two locations, A and B, each having a status of either 'Dirty' or 'Clean'.

Step 2: Initialize the vacuum agent's starting location.

Step 3: Sense the status (percept) of the current location.

Step 4: If the current location is dirty, perform the 'Suck' action to clean it.

Step 5: If the current location is clean, move the agent to the other location.

Step 6: Repeat Steps 3 to 5 for a fixed number of steps or until both locations become clean.

Step 7: Display the action taken and the environment status after every step.

Step 8: Stop when both Room A and Room B are clean.

Source Code:

```
# Vacuum Cleaner Problem using a Simple Reflex Agent

def vacuum_agent(environment, location):
    steps = 0
    while 'Dirty' in environment.values():
        steps += 1
        status = environment[location]
        print(f"Step {steps}: Agent at {location}, Status = {status}")

        if status == 'Dirty':
            print(f"Action: Suck (cleaning {location})")
            environment[location] = 'Clean'
        else:
            location = 'B' if location == 'A' else 'A'
            print(f"Action: Move to {location}")

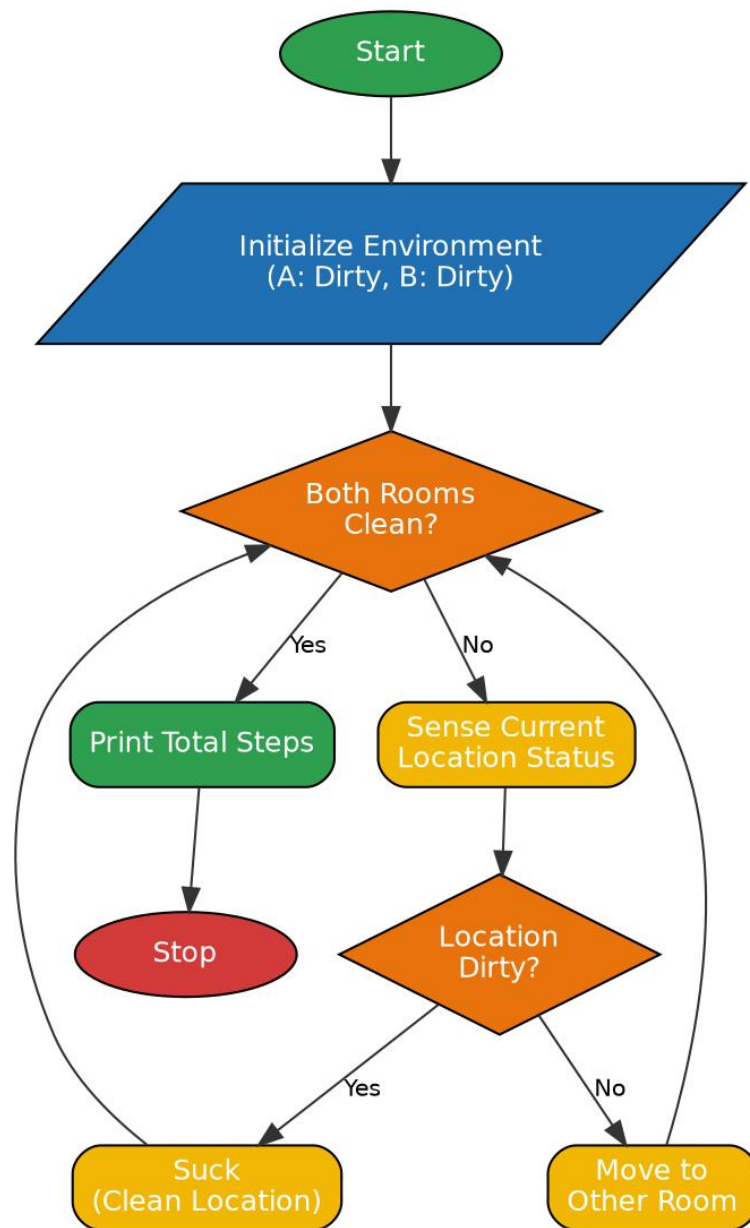
        print("Environment:", environment)
        print("-" * 30)

    print(f"Both rooms are clean. Total steps taken = {steps}")

environment = {'A': 'Dirty', 'B': 'Dirty'}
start_location = 'A'
```

```
print("Initial Environment:", environment)
vacuum_agent(environment, start_location)
```

Flowchart:



Output:

```
===== RESTART: C:/Users/yanar/Documents/CSA1717/EXP6.py =====
Initial Environment: {'A': 'Dirty', 'B': 'Dirty'}
Step 1: Agent at A, Status = Dirty
Action: Suck (cleaning A)
Environment: {'A': 'Clean', 'B': 'Dirty'}
-----
Step 2: Agent at A, Status = Clean
Action: Move to B
Environment: {'A': 'Clean', 'B': 'Dirty'}
-----
Step 3: Agent at B, Status = Dirty
Action: Suck (cleaning B)
Environment: {'A': 'Clean', 'B': 'Clean'}
-----
Both rooms are clean. Total steps taken = 3
>>> |
```

Result:

Thus the program for the Vacuum Cleaner Problem was written, executed successfully and the output was verified.